Does College Development Predict NBA Value? A Pre-Registered Walk-Forward Test on the 2014–21 Draft Classes

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Abstract

Does the way a college player improves from season to season tell a team anything about his NBA value that his final college season and the consensus big board do not? A specification frozen on August 10, 2026, before any NBA outcome was joined, set the test: add 17 development features to final-season college statistics, predict each draft class from 2014 to 2021 with models trained only on earlier classes, and require a mean gain of at least +0.02 in the Spearman correlation with 4-year regular-season VORP, higher in 5 of 8 classes, with a one-sided p below 0.05. Neither model met the bar. With gradient boosting, the pre-registered primary model, development changed the correlation by +0.01 (higher in 4 of 8 classes, one-sided p = 0.41), and with an elastic net by +0.02 (4 of 8, one-sided p = 0.33). The test rules out gains above about +0.10 (one-sided 95%), but at the +0.02 it targeted it had only 11–13% power, so a small gain is not ruled out. The consensus big board alone ranked prospects as well as the best model: a mean Spearman correlation of 0.46 (95% class bootstrap 0.39 to 0.53), against 0.47 for the elastic net with statistics and rank. The study covers prospects with two or more Division I seasons, which leaves out most top-10 picks. An October 2026 data validation found playoff rows mixed into the outcome and several wrong identity joins; the corrected rerun leaves the conclusion in place.

1. Question

Teams read college statistics two ways: as a level (how good a player was in his final season) and as a trajectory (how much he improved from one season to the next). The study asks whether the trajectory adds predictive information about a prospect's first four NBA seasons beyond his final-season level (H1), and beyond the consensus pre-draft big board (H2). It is a question about prediction. The design can show whether development improved out-of-sample ranking; it cannot show whether development matters to how players become valuable.

2. Data

Input	Source	Use
Draft results and NBA outcomes	Basketball-Reference draft pages and regular-season advanced tables	who was drafted where; 4-year regular-season VORP
College statistics and team strength	barttorvik.com player advanced statistics and team results, 2008–2026	final-season levels and season-to-season development
Recruiting rank and size	RSCI final rankings; NBA Draft Combine anthropometrics (archived pages)	levels baseline, with missing-value flags
Consensus rank	22 archived pre-draft media big boards (DraftExpress, NBADraft.net, The Stepien, Rookie Scale), 2,199 rankings	one or two boards per class; 2019 and 2020 rest on one board

The universe is 1,207 prospects from the 2010–21 classes: 720 drafted players and 487 who entered the draft from Division I and appeared on an archived board without being drafted. Development needs two seasons, so the primary population is the 893 with two or more Division I seasons. Classes 2010–13 are used only for training, which leaves 559 test players in the eight classes 2014–21 (53 to 80 per class).

The outcome is 4-year regular-season VORP: each regular season's VORP floored at zero and summed over the four seasons after the draft, with a player who never played counted as zero. 64.6% of test players are at exactly zero (median 0, maximum 14; Appendix, Figure A1), so the rank correlation mostly measures who reaches positive value.

> Who this covers: the 559 prospects of the 2014-21 classes with 2+ Division I seasons who were drafted or listed on an archived media big board; 91.2% of them ranked outside the board top 20, and only 17 of the 80 top-10 picks of those classes are among them. It says nothing about one-and-done, international or other non-Division I prospects, and its intervals reflect sampling noise, not who is in the sample.

3. Design

The specification (v1.0) was frozen in a commit on August 10, 2026, at 15:19; the first NBA outcome was joined five minutes later. Its research sections are unchanged since the freeze; later changes are logged as numbered amendments.

- **Feature sets.** B0: consensus board rank alone. B1, the levels baseline: 14 final-season college rates shrunk toward priors for sample size, plus age at draft, recruiting rank, height, weight and wingspan, team strength and minutes share. B2: B1 plus consensus rank. T, development: in each of 8 skill families (shooting, finishing, playmaking, ball security, defensive playmaking, rebounding, load, efficiency), the improvement beyond what a model of typical development

expects from the prior level, and a multi-season slope; plus the number of seasons (17 features).

- **Models.** Gradient boosting (primary) and an elastic net (secondary), each tuned inside its training window on a frozen grid by holding out the window's last class, then refit.
- **Walk-forward.** Each test class is predicted by models trained only on earlier classes, and every input is fixed by draft night (Figure 1).
- **Score and decision rule.** Within each class, the Spearman correlation between predicted and actual 4-year regular-season VORP. The H1 gain is that correlation for B1 + T minus B1, class by class. The frozen bar: a mean gain of at least +0.02, higher in at least 5 of 8 classes, and a one-sided exact sign-flip p below 0.05 over the eight class differences (all 256 sign patterns). The one-sided p is the share of sign patterns that give a mean gain at least as large: if development carried no predictive information, a gain this large would be that common.
- **H2.** A score built from development alone, against value beyond what consensus rank implies; and a redraft in which the model picks at each slot, against the players teams actually took.
- **Evidence types.** Every result is labeled: pre-registered (the boosting H1 test is the binding one), descriptive (comparisons with consensus rank, not pre-registered), or diagnostic and exploratory analyses added after results. The 2022 class is a holdout under Amendment #3.

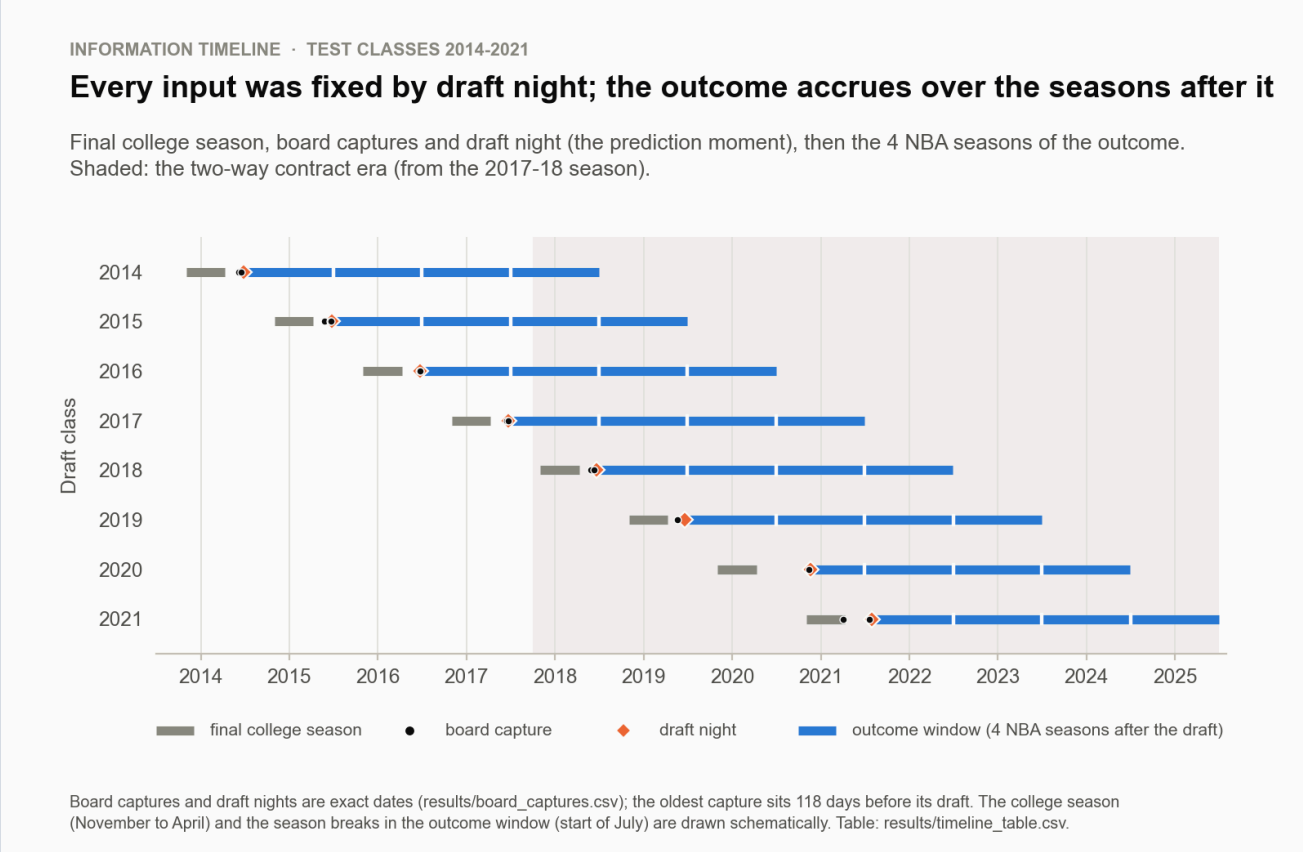


Figure 1. What was known by draft night. For each test class: the final college season, the board captures and draft night (the moment of prediction), then the four NBA seasons of the outcome. The oldest board capture sits 118 days before its draft. Shaded: the two-way contract era, from the 2017–18 season.

4. Results

4.1 Development did not improve prediction (H1, pre-registered)

Model	Mean gain	90% class bootstrap	Classes higher	One-sided p	One-sided 95% bound	Frozen bar
Gradient boosting (primary, binding)	+0.01	−0.047 to +0.055	4 of 8	0.41	+0.07	not met
Elastic net (secondary)	+0.02	−0.041 to +0.090	4 of 8	0.33	+0.10	not met

Table 1. Gain = Spearman correlation of statistics plus development minus statistics only, averaged over the eight test classes. The 90% interval is the one the specification defines (descriptive); the bound is a one-sided 95% t bound over the class differences.

Development did not improve out-of-sample prediction beyond final-season level: the gain was small and inconsistent across classes, and the test rules out gains above about +0.10 (elastic net) and +0.07 (boosting), one-sided 95%. That is a bound, not evidence of no gain. At the +0.02 the bar targeted, the test had 13% power with boosting and 11% with the elastic net. It reaches 80% power only for true gains of about +0.095 and +0.120, and an interval of ±0.02 would need 81 and 138 draft classes. Adding development to statistics and rank (the H2 model arm) changed the correlation by −0.01 with boosting (90% class bootstrap −0.044 to +0.020) and by +0.00 with the elastic net (−0.008 to +0.015).

4.2 The consensus board ranked prospects as well as the best model (descriptive)

Predictor (8-class mean Spearman)	Gradient boosting	Elastic net
Statistics only (B1)	0.29 (0.23 to 0.35)	0.40 (0.32 to 0.49)
Statistics + development (B1 + T)	0.29 (0.21 to 0.36)	0.42 (0.38 to 0.47)
Statistics + consensus rank (B2)	0.38 (0.34 to 0.43)	0.47 (0.40 to 0.53)
Consensus rank alone (B0, no model)	0.46 (0.39 to 0.53)	

Table 2. Mean over the classes 2014–21 of each predictor's Spearman correlation with 4-year regular-season VORP, with 95% class-bootstrap intervals. Every model was trained only on earlier classes.

Consensus rank alone matched the best model, the elastic net with statistics and rank: the model minus rank is +0.01 (95% class bootstrap −0.02 to +0.04, one-sided p = 0.36). Rank scored higher than the boosting models, by +0.08 over boosting with statistics and rank (95% class bootstrap +0.01 to +0.15, higher in 7 of 8 classes, one-sided p = 0.027) and by +0.17 over boosting with statistics only (+0.09 to +0.27, one-sided p = 0.008). With the 2018 class left out, the first gap is +0.05. These comparisons were not pre-registered, so they describe these eight classes rather than test a hypothesis (Figure 2).

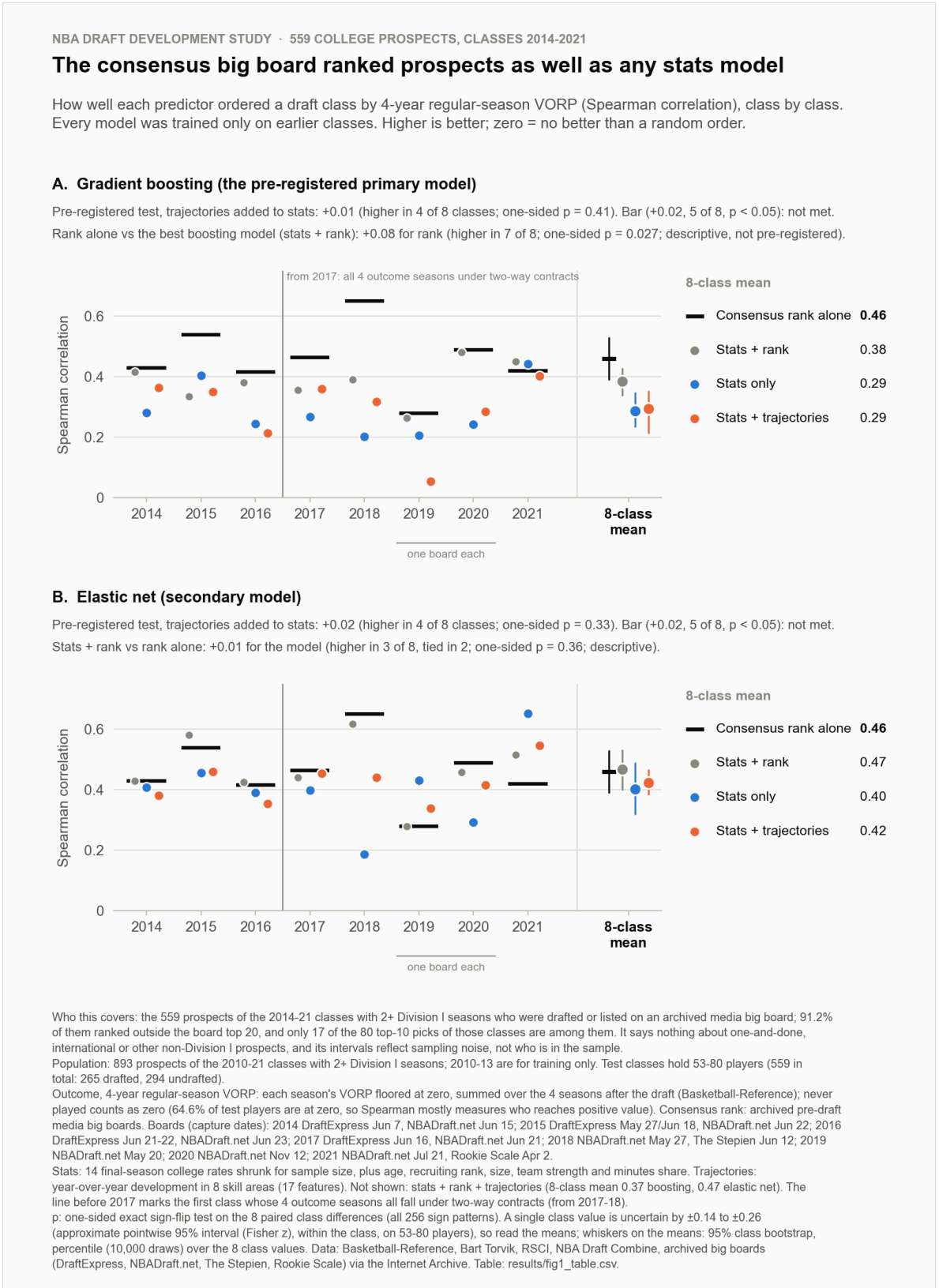


Figure 2. Spearman correlation of each predictor with 4-year regular-season VORP, class by class, with the 8-class means and their 95% class-bootstrap intervals; 559 prospects. A single class value is uncertain by ±0.14 to ±0.26, so read the means. Panel A: gradient boosting; panel B: the elastic net.

4.3 Development given the consensus (H2, pre-registered)

A score built from development alone did not predict value beyond what consensus rank implies: −0.04 (90% class bootstrap −0.102 to +0.018, one-sided p = 0.84). A redraft that let the model choose at each slot produced 32.0† less 4-year regular-season VORP than the players teams actually took (95% class bootstrap −51.6 to −15.8), lower in all eight classes (per-class median −3.70). Teams' actual picks out-produced the model's redraft in every class.

4.4 The middle of the draft (pre-registered diagnostic)

For the 241 test prospects ranked 21 to 60 by the consensus, the outcome is reaching 2,000 regular-season minutes within four seasons. Consensus rank resolves about 13%† of the outcome's uncertainty (95% player bootstrap 7% to 24%). The full model resolves 9%† and was no better, with a Brier score higher than rank's by +0.015 (95% player bootstrap −0.0067 to +0.0376). The middle of the draft is hard to call from public data.

4.5 The 2022 holdout

The 2022 class (58 players) was held out as a confirmatory check under Amendment #3, which substitutes the actual pick for board rank. It has been scored three times: on a superseded board, on the shipped board, and after retraining on the corrected labels. On the third scoring the declared projection's Spearman correlation with 4-year regular-season VORP is 0.493, against 0.465 for draft position alone: a difference of +0.028 (90% player bootstrap −0.019 to +0.077), a statistical tie. Because the construct uses the actual pick, its board is close to the draft by design (Spearman 0.983 with the actual pick).

4.6 Robustness (exploratory, added after results)

Two variants were added after results and are labeled exploratory. Coding rank on a log scale with an indicator for unboarded players leaves the comparisons as they are. A five-season outcome window gives an H1 gain of −0.04 with boosting and +0.02 with the elastic net, still short of the bar. The pre-registered battery of 30 one-sided comparisons (secondary outcomes and subsets) has none below 0.05; the smallest is 0.063.

5. Checks and diagnostics

- **Future-blindness gate.** Rebuilding the inputs as they stood at each draft gives identical rows at all 12 cuts.
- **Leakage self-tests.** A planted outcome reader and a planted web fetch are both caught before any model runs.
- **Shuffled outcomes.** Over 9 seeds with the outcome shuffled within class, the mean gain was −0.027 (SD 0.0435) and no seed met the bar. A one-sided p below 0.05 appeared in 1 of 18 reference runs, about what chance gives.
- **Second route.** The project's cross-check agrees in 4 of 4 independent re-computations. A separate re-implementation of the walk-forward, written from the frozen specification during

the October verification, reproduces every 8-class mean and both H1 gains to four decimals, and an independent parse of the outcome pages matches all 1,265 outcome values.

- **Tests.** 112 tests pass. They hold known answers taken from outside the code, and data-fix tests that fail on the shipped files.

6. Data corrections

A two-pass data validation on October 21–22, 2026 checked the datasets against independent sources. It found that the outcome builder read every table on each Basketball-Reference season page, so playoff rows replaced regular-season rows for many player-seasons and 198 of 1,268 four-year values changed. Seven drafted players listed under another name on a board were also counted as undrafted. Some identity joins were wrong (six wrong-person outcome joins, seven missed joins and three players kept in a class they did not enter), and the drafted-to-college crosswalk had gaps. Each fix was applied one at a time to a rebuild that first reproduced the shipped files exactly. Earlier, in August, a second review had found that every undrafted prospect was stored with zero value; 65 of them have positive 4-year regular-season VORP.

Quantity	As shipped	Corrected
Universe members (of them undrafted)	1,210 (490)	1,207 (487)
Test players, 2014–21	557	559
Total 4-year VORP, all outcome rows	914.9	1,103.2
Undrafted prospects with VORP above zero	62	65
Consensus rank alone, 8-class mean	0.4533	0.4603
H1 gain, boosting (classes higher; one-sided p)	−0.0346 (2 of 8; 0.96)	+0.0071 (4 of 8; 0.41)
H1 gain, elastic net	−0.0098 (3 of 8; 0.62)	+0.0221 (4 of 8; 0.33)
Redraft minus actual picks (VORP)	−20.8	−32.0

Table 3. The boosting gain changes sign between data versions, including fixes that touch only a few players; the frozen bar fails under every version, for both models.

Two earlier readings are withdrawn: that teams' drafting errors concentrate in the late draft (a product of how the redraft credits slots), and that the 2022 holdout was opened once (it was scored three times, and all three are reported).

7. Limitations

Limit	Mechanism	Likely direction
Opportunity	VORP piles up with minutes, and teams give high picks more minutes.	Favors consensus rank. A per-minute check (rank 0.129 against 0.212 for statistics only, 307 players) points the other way, but it exists only for players with 1,000+ minutes, a sample selected on opportunity.
Who is in the sample	The two-season rule removes one-and-done and most lottery prospects.	Restricts the range of talent and lowers every correlation; the result does not reach the top of the draft.
Noisy development	Development is a difference of two noisy, shrunk seasons.	Pulls the gain toward zero. Excess development correlates only +0.004 to +0.169 with itself across season pairs.
Overlap with levels	Development is built from the same seasons as the levels baseline.	Only what levels and rank miss can add, which shrinks the gain.
Four-season window	It misses late developers, the group a development hypothesis is about.	Pulls the gain toward zero. The five-season variant still falls short of the bar.
Two-way contracts	From 2017–18 they cap minutes for second-round and undrafted players, most of the test population.	Lowers measured value for late picks in later classes.
Media boards as the consensus	They stand in for team boards; two classes rest on one board.	A noisier consensus understates the true one, so "rank matched the best model" is conservative for rank.
Defense	College box scores and VORP both see defense poorly.	Unknown, and untested.
Eight test classes	The binding test has eight class differences.	A power limit, not a bias: small gains cannot be told from zero.

8. How a team could use it

As a caution on development stories: in these eight classes, a prospect's measured improvement from season to season did not help rank NBA value once his final season was known, and a stats model with the consensus rank did no better than the rank alone. The consensus board is the baseline a stats model should be judged against. Boards pool many evaluators and already absorb public college statistics, so a model is most useful where it disagrees with the board, as a question

for scouts. The result covers multi-year Division I players outside the top of the draft, and it cannot rule out a gain below about +0.10 in correlation.

9. Reproducibility and how this was built

One command, `./run_all.sh`, empties the results and figures folders and rebuilds everything from frozen inputs whose md5s are listed in `data/raw_manifest.csv`. It runs the data build with every fix, the tests and leakage checks before any model (including the 12-cut future-blindness gate), every pre-registered and robustness cell, the 2022 holdout, the analyses added after results, the cross-check, the fix ledger, the key numbers and every figure. It then checks every number in the documents against the results and runs a share scan. Each results table is stamped with a hash of its inputs, the seed and the data-fix set. Row-level data stays in a private folder outside the repository.

The records show the author's decisions: the frozen specification and its bar (August 10, 2026), the request for a second review after results existed (which found the undrafted zeros), Amendment #3 declaring the final model and opening the 2022 holdout, and acceptance of the October data validation's fixes and rerun. AI coding tools (Claude Code) wrote the code and drafted this paper from the project's results files, and AI review agents ran the reviews. The revision applied lessons from Syracuse coursework in statistics, sport analytics and information studies: nulls reported with their bound, every p-value labeled one-sided, verbs that match the design, intervals that resample whole draft classes, and identity checks on every join.

† Moved by the October 2026 data fixes and not yet reproduced by a second, independent route; every other headline number in sections 4.1 and 4.2 was reproduced by the independent re-implementation in section 5.

Data sources (capture dates)

- Basketball-Reference draft pages (2010–2022) and regular-season advanced tables (seasons 2011–2026), August 10, 2026; a third-party mirror of Basketball-Reference tables (files dated April 13, 2026), fetched May 4, 2026, used for independent checks.
- barttorvik.com player advanced statistics and team results, 2008–2026, August 10–11, 2026.
- RSCI final rankings, August 10–11, 2026. NBA Draft Combine anthropometrics via Internet Archive captures of stats.nba.com, August 10, 2026.
- Pre-draft big boards (DraftExpress, NBADraft.net, The Stepien, Rookie Scale) via the Internet Archive, captured August 10, 2026; board capture dates by class are in the repository.

Appendix

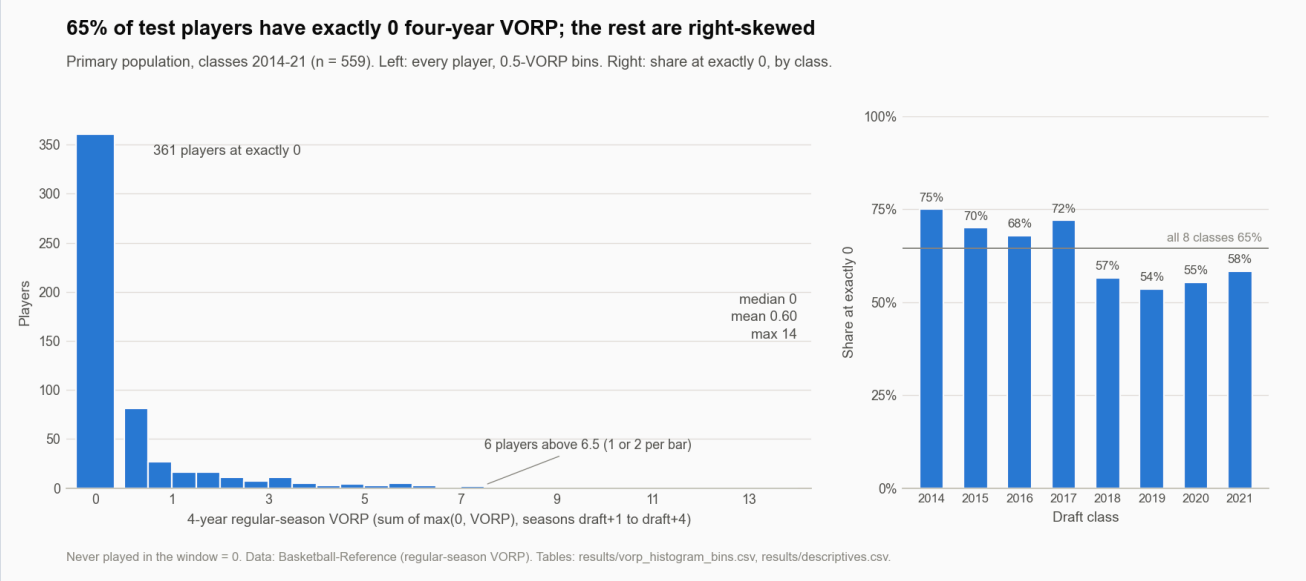


Figure A1. The outcome. Left: 4-year regular-season VORP of the 559 test players in 0.5-VORP bins. Right: the share at exactly zero, by class.

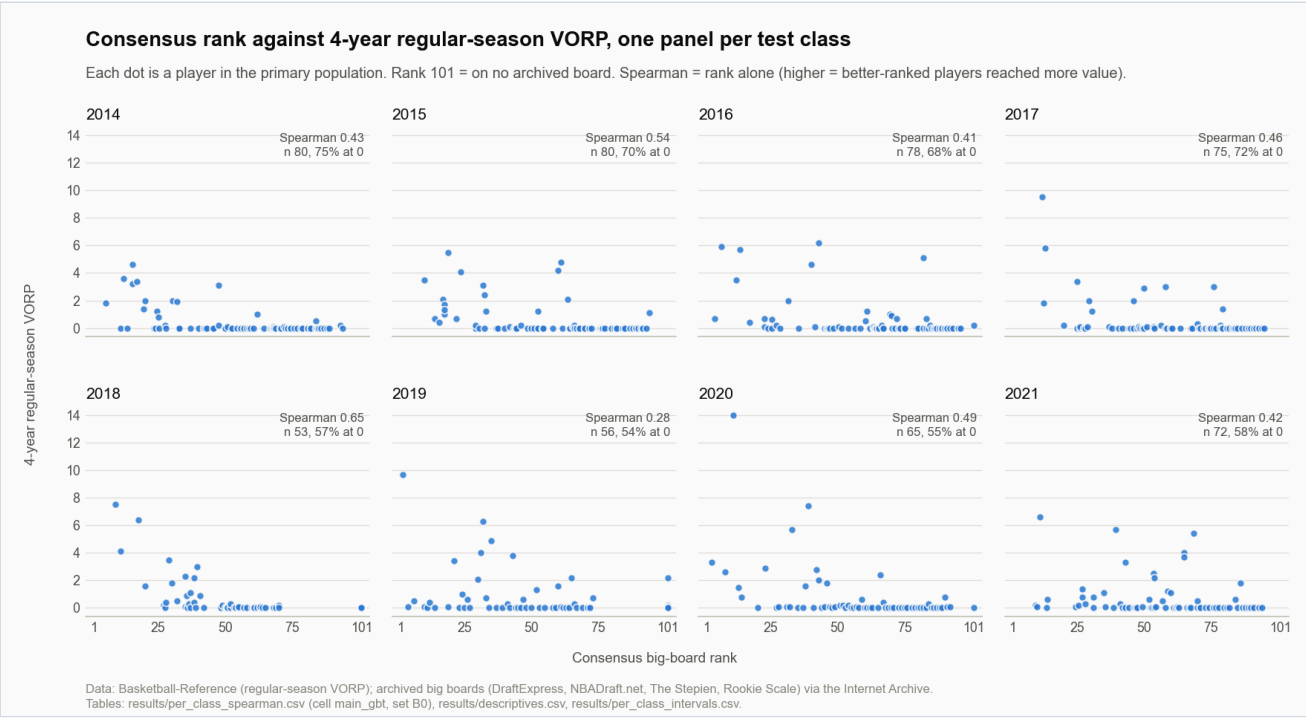


Figure A2. Consensus rank against 4-year regular-season VORP, one panel per test class, with the Spearman correlation of rank alone. Rank 101 = on no archived board.